Applied

Calculus IV

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- Some examples and content in this PPT are adapted from the reference textbook (Differential Equation by Dennis G. Zill). Full credit for the original textbook material belongs to its respective author and publisher.

Week 4 & 5

- Review
- 2.4
- 2.5

Contents

First-Order Differential Equations

Recall

Differential of $f(x,y)$

(a function of two variables with continuous first partial derivatives in a region R of the *xy*-plane)

$$z = f(x,y) \rightarrow dz = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

$$c = f(x,y) \rightarrow 0 = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

Example:

$$x^2 + 2xy + y^3 = c \quad \rightarrow \quad (2x + 2y)dx + (2x + 3y^2)dy = 0$$

$$e^x - y + 2xy = c \quad \rightarrow \quad (e^x + 2y)dx + (2x - 1)dy = 0$$

$$x^3 - ye^x + 2x\, siny = c$$

$$\rightarrow (3x^2 - ye^x + 2\, siny)dx + (-e^x + 2x\, cosy)dy = 0$$

Question: Can we do it reversely?

(can we find the solution of a differential equation)

Solution		**DE**
$x^2 + 2xy + y^3 = c$	$\Leftarrow$	$(2x + 2y)dx + (2x + 3y^2)dy = 0$
$e^x - y + 2xy = c$	$\Leftarrow$	$(e^x + 2y)dx + (2x - 1)dy = 0$

Let us write our DE in this form : $M(x,y)dx + N(x,y)dy = 0$

If we could find a function $f(x, y)$ such that $M(x,y) = \dfrac{\partial f}{\partial x}$ & $N(x,y) = \dfrac{\partial f}{\partial y}$

obviously, $f(x, y) = c$ *is* the solution of the DE.

$$\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0$$

Note that not every first-order DE written in differential form $M(x, y)\, dx + N(x, y)\, dy = 0$ corresponds to a differential of $f(x, y) = c$.

DEFINITION 2.4.1 Exact Equation

A differential expression $M(x, y)\,dx + N(x, y)\,dy$ is an **exact differential** in a region R of the xy-plane if it corresponds to the differential of some function $f(x, y)$ defined in R. A first-order differential equation of the form

$$M(x, y)\,dx + N(x, y)\,dy = 0$$

is said to be an **exact equation** if the expression on the left-hand side is an exact differential.

* Not every first-order DE is an exact differential equation.

Example

$$(2x + 2y)dx + (2x + 3y^2)dy = 0$$

$$\frac{\partial f}{\partial x} = 2x + 2y \quad \& \quad \frac{\partial f}{\partial y} = 2x + 3y^2$$

$$\Rightarrow f(x, y) = x^2 + 2xy + y^3$$

Since we could find $f(x, y) = c$, this is an exact differential equation

Recall

If f_x and f_y are continuous, f_{xy} and f_{xy} exist, and are continuous in a rectangular region R

$$f_{xy} = f_{yx}$$

or $$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

From the other hand,

$$M(x, y)dx + N(x, y)dy = 0 \quad \Rightarrow \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$M(x,y) = \frac{\partial f}{\partial x} \qquad N(x,y) = \frac{\partial f}{\partial y}$$

THEOREM 2.4.1 Criterion for an Exact Differential

Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first partial derivatives in a rectangular region R defined by $a < x < b$, $c < y < d$. Then a necessary and sufficient condition that $M(x, y)\,dx + N(x, y)\,dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \tag{4}$$

Method of Solution?

Assume it is an exact DE

$$M_y = N_x$$

$$M(x,y)dx + N(x,y)dy = 0$$

$$= \frac{\partial f}{\partial x} \qquad = \frac{\partial f}{\partial y}$$

$$f(x,y) = \int M(x,y)dx + g(y)$$

$$\Rightarrow \frac{\partial f(x,y)}{\partial y} = \frac{\partial}{\partial y}\int M(x,y)dx + \frac{\partial}{\partial y}g(y) = N(x,y) \Rightarrow \frac{\partial}{\partial y}g(y) = N(x,y) - \frac{\partial}{\partial y}\int M(x,y)dx$$

$$g(y) = \int \left(N(x,y) - \frac{\partial}{\partial y}\int M(x,y)dx \right) dy$$

$$M(x,y)dx + N(x,y)dy = 0$$

$$\downarrow \qquad\qquad \downarrow$$

$$\frac{\partial f}{\partial x} \qquad\qquad \frac{\partial f}{\partial y}$$

Alternatively,

$$f(x,y) = \int N(x,y)dy + h(x)$$

$$\Rightarrow \frac{\partial f(x,y)}{\partial x} = \frac{\partial}{\partial x}\int N(x,y)dy + \frac{\partial}{\partial x}h(x) = M(x,y) \Rightarrow \frac{\partial}{\partial x}h(x) = M(x,y) - \frac{\partial}{\partial x}\int N(x,y)dy$$

$$h(x) = \int\left(M(x,y) - \frac{\partial}{\partial x}\int N(x,y)dy\right)dx$$

Summary

If $M_y = N_x$

Step 1. $$f(x,y) = \int M(x,y)dx + g(y)$$

Step 2. $$g(y) = \int \left(N(x,y) - \frac{\partial}{\partial y} \int M(x,y)dx \right) dy$$

Step 1. $$f(x,y) = \int N(x,y)dy + h(x)$$

Step 2. $$h(x) = \int \left(M(x,y) - \frac{\partial}{\partial x} \int N(x,y)dy \right) dx$$

Depending on the difficulty of integrations, you can choose one way

Example: Slove $$2xydx + (x^2 - 1)dy = 0$$

$$\underset{M(x,y)}{2xy}\,dx + \underset{N(x,y)}{(x^2-1)}\,dy = 0$$

$M_y = N_x = 2x$ ✓

$$f(x,y) = \int M(x,y)dx + g(y) = \int 2xydx + g(y) = yx^2 + g(y)$$

$$\frac{\partial f(x,y)}{\partial y} = x^2 + \frac{d}{dy}g(y) = N(x,y) = x^2 - 1 \Rightarrow \frac{\partial}{\partial y}g(y) = x^2 - 1 - x^2$$

$$\Rightarrow \quad g(y) = -y$$

$$\Rightarrow \quad f(x,y) = yx^2 - y$$

$$\Rightarrow \quad yx^2 - y = c \Rightarrow \quad y = \frac{c}{x^2 - 1}$$ defined on $(1, \infty)$

Example: Slove $\frac{dy}{dx} = -\frac{2xy}{x^2+3y^2+1}$ $\longrightarrow$ $2xydx + (x^2 + 3y^2 + 1)dy = 0$

$M(x,y)$ $N(x,y)$

$M_y = N_x = 2x$ ✓

$$f(x,y) = \int M(x,y)dx + g(y) = \int 2xydx + g(y) = yx^2 + g(y)$$

$$\frac{\partial f(x,y)}{\partial y} = x^2 + \frac{d}{dy}g(y) = N(x,y) = x^2 + 3y^2 + 1 \Rightarrow \frac{\partial}{\partial y}g(y) = x^2 + 3y^2 + 1 - x^2$$

$$\Rightarrow \quad g(y) = y^3 + y$$

$$\Rightarrow \quad f(x,y) = yx^2 + y^3 + y$$

$$\Rightarrow \quad y(x^2+1) + y^3 = c$$

Example: Slove $\frac{dy}{dx} = -\frac{2xy}{x^2+3y^2+1}$ $\longrightarrow$ $\underbrace{2xy}_{M(x,y)}dx + (\underbrace{x^2+3y^2+1}_{N(x,y)})dy = 0$

$M_y = N_x = 2x$ ✓

$$f(x,y) = \int M(x,y)dx + g(y) = \int 2xydx + g(y) = yx^2 + g(y)$$

$$\frac{\partial f(x,y)}{\partial y} = x^2 + \frac{\partial}{\partial y}g(y) = N(x,y) = x^2 + 3y^2 + 1 \Rightarrow \frac{\partial}{\partial y}g(y) = x^2 + 3y^2 + 1 - x^2$$

$$\Rightarrow \quad g(y) = y^3 + y$$

$$\Rightarrow \quad f(x,y) = yx^2 + y^3 + y$$

$$\Rightarrow \quad y(x^2+1) + y^3 = c$$

Exercises 2.4

In Problems 1–20 determine whether the given differential equation is exact. If it is exact, solve it.

1. $(2x - 1)\,dx + (3y + 7)\,dy = 0$

2. $(2x + y)\,dx - (x + 6y)\,dy = 0$

3. $(5x + 4y)\,dx + (4x - 8y^3)\,dy = 0$

4. $(\sin y - y \sin x)\,dx + (\cos x + x \cos y - y)\,dy = 0$

5. $(2xy^2 - 3)\,dx + (2x^2y + 4)\,dy = 0$

6. $\left(2y - \dfrac{1}{x} + \cos 3x\right)\dfrac{dy}{dx} + \dfrac{y}{x^2} - 4x^3 + 3y \sin 3x = 0$

7. $(x^2 - y^2)\,dx + (x^2 - 2xy)\,dy = 0$

8. $\left(1 + \ln x + \dfrac{y}{x}\right)dx = (1 - \ln x)\,dy$

9. $(x - y^3 + y^2 \sin x)\,dx = (3xy^2 + 2y \cos x)\,dy$

10. $(x^3 + y^3)\,dx + 3xy^2\,dy = 0$

1. Solve $(2x-1)dx + (3y+7)dy = 0$

$M(x,y)$ $N(x,y)$

$M_y = N_x = 0$ ✓

$$f(x,y) = \int (2x-1)dx + g(y) = x^2 - x + g(y)$$

$$\frac{\partial f(x,y)}{\partial y} = g'(y) = N(x,y) = 3y+7$$

$$\Rightarrow \quad g' = 3y+7 \quad \Rightarrow \quad g(y) = \frac{3}{2}y^2 + 7y$$

$$x^2 - x + \frac{3}{2}y^2 + 7y = c$$

4. $(\sin y - y\sin x)\,dx + (\cos x + x\cos y - y)\,dy = 0$

$M(x,y)$ $N(x,y)$

$M_y = N_x = \cos y - \sin x$ ✓

$$f(x,y) = \int (\sin y - y\sin x)\,dx + g(y) = x\sin y + y\cos x + g(y)$$

$$\frac{\partial f(x,y)}{\partial y} = x\cos y + \cos x + g'(y) = N(x,y) = \cos x + x\cos y - y$$

$$\Rightarrow \quad g' = \cos x + x\cos y - y - x\cos y + \cos x = -y$$

$$\Rightarrow \quad g(y) = -\tfrac{1}{2}y^2$$

$$x\sin y + y\cos x - \frac{1}{2}y^2 = c$$

7. Solve $(x^2 - y^2)dy + (x^2 - 2xy)dx = 0$

$M(x, y)$ $N(x, y)$

$M_y \neq N_x$ The equation is not an exact differential equation

In Problems 21–26 solve the given initial-value problem.

21. $(x + y)^2\,dx + (2xy + x^2 - 1)\,dy = 0, \quad y(1) = 1$

22. $(e^x + y)\,dx + (2 + x + ye^y)\,dy = 0, \quad y(0) = 1$

23. $(4y + 2t - 5)\,dt + (6y + 4t - 1)\,dy = 0, \quad y(-1) = 2$

24. $\left(\dfrac{3y^2 - t^2}{y^5}\right)\dfrac{dy}{dt} + \dfrac{t}{2y^4} = 0, \quad y(1) = 1$

25. $(y^2 \cos x - 3x^2y - 2x)\,dx$

$+ (2y \sin x - x^3 + \ln y)\,dy = 0, \quad y(0) = e$

26. $\left(\dfrac{1}{1 + y^2} + \cos x - 2xy\right)\dfrac{dy}{dx} = y(y + \sin x), \quad y(0) = 1$

Similar to the previous exercises (and at the end, find the value of parameter c by using the initial condition)

Challange: What if $M_y \neq N_x$?

We search for an Integrating Factors $\mu(x,y)M(x,y)dx + \mu(x,y)N(x,y)dy = 0$

If it becomes an exact differential, then

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x$$

$$\mu(M_y - N_x) = \mu_x N - \mu_y M$$

if $\mu_x = 0$

$$\frac{d\mu}{dy} = -\frac{M_y - N_x}{M}\mu$$

if $\mu_y = 0$

$$\frac{d\mu}{dx} = \frac{M_y - N_x}{N}\mu$$

Conclusion (integrating factor for non-exact DE)

If $\frac{N_x - M_y}{M}$ is a function of y alone, $\mu(y) = e^{\int \frac{N_x - M_y}{M}}$ Why?

If $\frac{M_y - N_x}{N}$ is a function of x alone, $\mu(x) = e^{\int \frac{M_y - N_x}{N}}$ Why?

Example: Slove $xydx + (2x^2 + 3y^2 - 20)dy = 0$

$M_y \neq N_x$ $\quad\quad$ $M_y - N_x = x - 4x = -3x$

$$\frac{M_y - N_x}{N} = \frac{-3x}{2x^2 + 3y^2 - 20}$$ is not a function of x alone

$$\frac{N_x - M_y}{M} = \frac{3x}{xy} = \frac{3}{y}$$ is a function of y alone $\Rightarrow$ $\mu(y) = e^{\int \frac{3}{y}} = y^3$ Integrating Factor

$$\Rightarrow \quad xy^4dx + (2x^2y^3 + 3y^5 - 20y^3)dy = 0$$

Now $\quad xy^4dx + (2x^2y^3 + 3y^5 - 20y^3)dy = 0$ is an exact DE

$M(x,y) \qquad N(x,y) \qquad M_y = N_x = 4x\,y^3$ ✓

$$f(x,y) = \int M(x,y)dx + g(y) = \int xy^4dx + g(y) = \frac{1}{2}x^2y^4 + g(y)$$

$$\frac{\partial f(x,y)}{\partial y} = 2x^2y^3 + g' = N(x,y) = 2x^2y^3 + 3y^5 - 20y^3 \Rightarrow \quad g' = 2x^2y^3 + 3y^5 - 20y^3 - 2x^2y^3$$

$$\Rightarrow \quad g(y) = \frac{3}{6}y^6 - 5y^4$$

$$\Rightarrow \quad f(x,y) = \frac{1}{2}x^2y^4 + \frac{3}{6}y^6 - 5y^4$$

$$\Rightarrow \frac{1}{2}x^2y^4 + \frac{1}{3}y^6 - 5y^4 = c$$

In Problems 27 and 28 find the value of k so that the given differential equation is exact.

27. $(y^3 + kxy^4 - 2x)\,dx + (3xy^2 + 20x^2y^3)\,dy = 0$

28. $(6xy^3 + \cos y)\,dx + (2kx^2y^2 - x\sin y)\,dy = 0$

$$M_y = N_x$$

27.

$$M_y = 3y^2 + 4kxy^3$$

$$N_x = 3y^2 + 40xy^3$$

$$3y^2 + 4kxy^3 = 3y^2 + 40xy^3$$

$$k = 10$$

28.

$$M_y = 18xy^2 - \sin y$$

$$N_x = 4kxy^2 - \sin y$$

$$18xy^2 - \sin y = 4kxy^2 - \sin y$$

$$k = \frac{9}{2}$$

In Problems 31–36 solve the given differential equation by finding, as in Example 4, an appropriate integrating factor.

31. $(2y^2 + 3x)\,dx + 2xy\,dy = 0$

32. $y(x + y + 1)\,dx + (x + 2y)\,dy = 0$

33. $6xy\,dx + (4y + 9x^2)\,dy = 0$

34. $\cos x\,dx + \left(1 + \dfrac{2}{y}\right)\sin x\,dy = 0$

35. $(10 - 6y + e^{-3x})\,dx - 2\,dy = 0$

36. $(y^2 + xy^3)\,dx + (5y^2 - xy + y^3 \sin y)\,dy = 0$

31. $(2y^2 + 3x)\,dx + 2xy\,dy = 0$

$M_y = 4y$ & $N_x = 2y$

$\frac{M_y - N_x}{N} = \frac{2y}{2xy} = \frac{1}{x} \rightarrow \mu(x) = e^{\int \frac{1}{x}} = x \rightarrow (2xy^2 + 3x^2)dx + 2x^2ydy = 0$

$f_x = 2xy^2 + 3x^2$

$f = x^2y^2 + x^3 + h(y)$

$h'(y) = 0$

$h(y) = 0$

Solution: $x^2y^2 + x^3 = c$

32. $y(x + y + 1)\,dx + (x + 2y)\,dy = 0$

$$M_y = x + 2y + 1 \quad \& \quad N_x = 1$$

$$\frac{M_y - N_x}{N} = \frac{x + 2y}{x + 2y} = 1 \rightarrow \mu(x) = e^{\int dx} = e^x \rightarrow \quad y(x+y+1)e^x dx + \underset{\underset{f_y}{\downarrow}}{(x+2y)e^x}\, dy = 0$$

$$f = xye^x + y^2e^x + h(x)$$

$$f_x = ye^x + xye^x + y^2e^x + h'(x)$$

$$h'(x) = 0$$

$$h(x) = 0$$

Solution: $xye^x + y^2e^x = c$

In Problems 37 and 38 solve the given initial-value problem by finding, as in Example 4, an appropriate integrating factor.

37. $x\,dx + (x^2y + 4y)\,dy = 0, \quad y(4) = 0$

38. $(x^2 + y^2 - 5)\,dx = (y + xy)\,dy, \quad y(0) = 1$

37. $x\,dx + (x^2y + 4y)\,dy = 0, \quad y(4) = 0$

$$M_y = 0 \quad \& \quad N_x = 2xy$$

$$\frac{M_y - N_x}{N} = \frac{-2xy}{x^2y + 4y} = \frac{-2x}{x^2+4} \rightarrow \mu(x) = e^{\int \frac{-2x}{x^2+4}dx} = e^{-\ln|x^2+4|} = \frac{1}{x^2+4}$$

$$\frac{x}{x^2+4}dx + y\,dy = 0 \rightarrow$$

$$\downarrow$$

$$f_x$$

$$f = \frac{1}{2}\ln(4+x^2) + h(y)$$

$$h'(y) = y$$

$$h(y) = \frac{1}{2}y^2$$

Solution: $\frac{1}{2}\ln(4+x^2) + \frac{1}{2}y^2 = c$

$$\ln(4+x^2) + y^2 = 2c$$

$$\ln(4+x^2) + y^2 = c_1$$

$$e^{\ln(4+x^2)+y^2} = e^{c_1}$$

$$e^{\ln(4+x^2)} \cdot e^{y^2} = e^{c_1}$$

$$(4+x^2)e^{y^2} = c_2$$

$$c_2 = 20$$

$$(4+x^2)e^{y^2} = 20$$

37. $x\,dx + (x^2y + 4y)\,dy = 0, \quad y(4) = 0$

❖ Another integrating factor for this DE

$M_y = 0 \quad \& \quad N_x = 2xy$

$$\frac{N_x - M_y}{M} = \frac{2xy}{x} = 2y \rightarrow \mu(x) = e^{\int 2y dy} = e^{y^2} \rightarrow \quad xe^{y^2}\,dx + y(x^2+4)e^{y^2}\,dy = 0$$

$\downarrow$

f_x

$$f = \int xe^{y^2}\,dx + g(y) = \frac{1}{2}x^2e^{y^2} + g(y)$$

$$f_y = \frac{1}{2}x^2(2ye^{y^2}) + g'(y) = yx^2e^{y^2} + 4ye^{y^2}$$

$$\rightarrow \quad g'(y) = 4ye^{y^2} \quad \rightarrow \quad g = 2e^{y^2}$$

$$f = \frac{1}{2}x^2e^{y^2} + 2e^{y^2} \quad \rightarrow \frac{1}{2}x^2e^{y^2} + 2e^{y^2} = c_1 \rightarrow x^2e^{y^2} + 4e^{y^2} = 4c_1 = c \rightarrow (x^2+4)e^{y^2} = c$$

$$y(4) = 0 \quad \rightarrow \quad c = 20$$

$$\rightarrow (x^2 + 4)e^{y^2} = 20$$

First-Order Differential Equations

2.1	Solution Curves Without a Solution
2.2	Separable Equations
2.3	Linear Equations
2.4	Exact Equations
2.5	Solutions by Substitutions
2.6	A Numerical Method

Simplify DE by substituting y (with your guess)

Example: Slove $x\frac{dy}{dx}+y=x^2y^2 \longrightarrow \frac{dy}{dx}+\frac{y}{x}=xy^2$

$$u=y^{-1} \rightarrow y=u^{-1} \rightarrow \frac{dy}{dx}=-u^{-2}\frac{du}{dx}$$

$$\rightarrow -u^{-2}\frac{du}{dx}+\frac{u^{-1}}{x}=xu^{-2} \rightarrow \frac{du}{dx}-\frac{u}{x}=-x$$

$$P(x)=\frac{-1}{x} \rightarrow e^{-\int\frac{1}{x}dx}=x^{-1} \rightarrow -\frac{1}{x}\frac{du}{dx}+\frac{u}{x^2}=-1 \rightarrow \frac{d}{dx}\left[\frac{u}{x}\right]=-1$$

$$\rightarrow \frac{u}{x}=-x+c \rightarrow u=-x^2+cx \rightarrow y^{-1}=-x^2+cx \rightarrow y=\frac{1}{-x^2+cx}$$

Example: Slove $\frac{dy}{dx} = (-2x + y)^2 - 7,\ \ y(0) = 0$

$$u = -2x + y \;\rightarrow\; \frac{du}{dx} = -2 + \frac{dy}{dx} \;\rightarrow\; \frac{du}{dx} + 2 = u^2 - 7$$

$$\frac{du}{dx} = u^2 - 9 \rightarrow \frac{du}{u^2 - 9} = dx \rightarrow \frac{1}{6}\left[\frac{1}{u-3} - \frac{1}{u+3}\right] du = dx$$

$$\frac{1}{6}\ln\frac{u-3}{u+3} = x + c \rightarrow \frac{u-3}{u+3} = e^{6x+6c} = c_1 e^x \rightarrow \frac{-2x+y-3}{-2x+y+3} = c_1 e^x \rightarrow c_1 = -1$$

$$\Rightarrow \frac{-2x + y - 3}{-2x + y + 3} = -e^x$$

homogeneous function of degree α

$$f(tx,ty) = t^{\alpha} f(x,y)$$

Example

$$f(x,y) = x^3 + y^3$$

homogeneous function of degree 3

$$f(tx,ty) = (tx)^3 + (ty)^3 = t^3(x^3+y^3) = t^3 f(x,y)$$

This first order differential equation is called homogeneous if M and N both are homogeneous functions of the same degree

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

In Problems 1–10 solve the given differential equation by using an appropriate substitution.

1. $(x - y)\,dx + x\,dy = 0$

2. $(x + y)\,dx + x\,dy = 0$

3. $x\,dx + (y - 2x)\,dy = 0$

4. $y\,dx = 2(x + y)\,dy$

5. $(y^2 + yx)\,dx - x^2\,dy = 0$

6. $(y^2 + yx)\,dx + x^2\,dy = 0$

7. $\dfrac{dy}{dx} = \dfrac{y - x}{y + x}$

8. $\dfrac{dy}{dx} = \dfrac{x + 3y}{3x + y}$

9. $-y\,dx + \left(x + \sqrt{xy}\right)dy = 0$

10. $x\dfrac{dy}{dx} = y + \sqrt{x^2 - y^2}, \quad x > 0$

Note that these equations are homogenous

and $y = ux$ usually is an appropriate substitution

1. $(x - y)dx + xdy = 0$

$y = ux \rightarrow dy = udx + xdu \rightarrow$

$$(x - ux)\,dx + x(u\,dx + x\,du) = 0$$

$$dx + x\,du = 0$$

$$\frac{dx}{x} + du = 0$$

$$\ln|x| + u = c$$

$$x\ln|x| + y = cx$$

2. $(x+y)dx + xdy = 0$

$y = ux \quad \rightarrow \quad dy = udx + xdu \quad \rightarrow \quad (x+ux)dx + x(udx + xdu) = 0$

$$(1+u)dx + udx + xdu = 0$$

$$(1+2u)dx + xdu = 0$$

$$\frac{du}{1+2u} + \frac{dx}{x} = 0$$

$$\frac{1}{2} ln|1+2u| + ln\,|x| = c_1$$

$$ln|1+2u| + 2\,ln|x| = 2c_1 \quad \rightarrow \quad (1+2u)x^2 = c$$

$$x^2 + 2x^2\frac{y}{x} = c$$

$$x^2 + 2xy = c$$

3. $xdx + (y - 2x)dy = 0$

$y = ux \rightarrow \quad dy = udx + xdu \quad \rightarrow \qquad xdx + (ux - 2x)(udx + xdu) = 0$

$$dx + (u - 2)(udx + xdu) = 0$$

$$(1 + u^2 - 2u)dx + x(u - 2)du = 0$$

$$\frac{(u-2)du}{1 + u^2 - 2u} + \frac{dx}{x} = 0 \rightarrow \frac{u-1}{(u-1)^2}du + \frac{-1}{(u-1)^2}du + \frac{dx}{x} = 0$$

$$\rightarrow \ln|u - 1| + \frac{1}{u-1} + \ln|x| = c_1$$

$$\ln|(u-1)x| + \frac{1}{u-1} = c_1 \rightarrow \ln\left|\left(\frac{y}{x} - 1\right)x\right| + \frac{1}{\frac{y}{x} - 1} = c_1$$

$$\rightarrow \ln|y - x| + \frac{x}{y - x} = c_1$$

3. $xdx + (y - 2x)dy = 0$

$x = vy \rightarrow$ $dx = vdy + ydv \rightarrow$

$$vy(v\,dy + y\,dv) + (y - 2vy)\,dy = 0$$

$$vy^2\,dv + y\left(v^2 - 2v + 1\right)dy = 0$$

❖ Alternatively, we can substitue x

$$\frac{v\,dv}{(v-1)^2} + \frac{dy}{y} = 0$$

$$\ln|v-1| - \frac{1}{v-1} + \ln|y| = c$$

$$\ln\left|\frac{x}{y} - 1\right| - \frac{1}{x/y - 1} + \ln|y| = c$$

$$\rightarrow \ln|x-y| - \frac{y}{x-y} = c_1$$

9. $-ydx + (x + \sqrt{xy})dy = 0$

$$y = ux \quad \rightarrow \quad dy = udx + xdu \quad \rightarrow -ux\,dx + (x + \sqrt{u}\,x)(u\,dx + x\,du) = 0$$

$$(x^2 + x^2\sqrt{u}\,)\,du + xu^{3/2}\,dx = 0$$

$$\left(u^{-3/2} + \frac{1}{u}\right)du + \frac{dx}{x} = 0$$

$$-2u^{-1/2} + \ln|u| + \ln|x| = c$$

$$\ln|y/x| + \ln|x| = 2\sqrt{x/y} + c$$

$$y(\ln|y| - c)^2 = 4x.$$

10. $x\dfrac{dy}{dx} = y + \sqrt{x^2 - y^2}, \qquad x > 0$

After rearrangment

$y = ux \;\rightarrow\; dy = u\,dx + x\,du \;\rightarrow$

$$\sqrt{x^2 - u^2x^2}\,dx - x^2\,du = 0$$

$$x\sqrt{1-u^2}\,dx - x^2\,du = 0, \qquad (x > 0)$$

$$\frac{dx}{x} - \frac{du}{\sqrt{1-u^2}} = 0$$

$$\ln x - \sin^{-1} u = c$$

$$\sin^{-1} u = \ln x + c_1$$

$$\sin^{-1}\frac{y}{x} = \ln x + c_2$$

$$\frac{y}{x} = \sin\left(\ln x + c_2\right)$$

$\rightarrow \quad y = x\sin\left(\ln x + c_2\right)$

In Problems 11–14 solve the given initial-value problem.

11. $xy^2 \dfrac{dy}{dx} = y^3 - x^3, \quad y(1) = 2$

12. $(x^2 + 2y^2) \dfrac{dx}{dy} = xy, \quad y(-1) = 1$

13. $(x + ye^{y/x})\, dx - xe^{y/x}\, dy = 0, \quad y(1) = 0$

14. $y\, dx + x(\ln x - \ln y - 1)\, dy = 0, \quad y(1) = e$

11. $xy^2 \frac{dy}{dx} = y^3 - x^3, \quad y(1) = 2$

$y = ux \quad \rightarrow \quad dy = udx + xdu \quad \rightarrow \quad \left(x^3 - u^3x^3\right) dx + u^2x^3(u\,dx + x\,du) = 0$

$$dx + u^2x\,du = 0$$

You also can replace $\frac{dy}{dx}$ as

$$\frac{dy}{dx} = x\frac{du}{dx} + u$$

$$\frac{dx}{x} + u^2\,du = 0$$

$$\ln|x| + \frac{1}{3}u^3 = c$$

$$3x^3 \ln|x| + y^3 = c_1x^3$$

$y(1) = 2 \quad \rightarrow \quad 3x^3 \ln|x| + y^3 = 8x^3$

13. $(x + ye^{y/x})\, dx - xe^{y/x}\, dy = 0, \quad y(1) = 0$

$y = ux \;\rightarrow\; dy = udx + xdu \;\rightarrow\; (x + uxe^u)\, dx - xe^u(u\, dx + x\, du) = 0$

$$dx - xe^u\, du = 0$$

$$\frac{dx}{x} - e^u\, du = 0$$

$$\ln|x| - e^u = c$$

$$\ln|x| - e^{y/x} = c.$$

$y(1) = 0 \;\rightarrow\; 0 - 1 = c \;\rightarrow c = -1$

In Problems 15–20 solve the given differential equation by using an appropriate substitution.

15. $x\dfrac{dy}{dx} + y = \dfrac{1}{y^2}$

16. $\dfrac{dy}{dx} - y = e^x y^2$

17. $\dfrac{dy}{dx} = y(xy^3 - 1)$

18. $x\dfrac{dy}{dx} - (1 + x)y = xy^2$

19. $t^2\dfrac{dy}{dt} + y^2 = ty$

20. $3(1 + t^2)\dfrac{dy}{dt} = 2ty(y^3 - 1)$

15. $x\frac{dy}{dx} + y = \frac{1}{y^2}$ $\rightarrow$ $\frac{dy}{dx} + \frac{1}{x}y = \frac{1}{x}y^{-2}$

$w = y^3 \rightarrow$ $dw = 3y^2dy \rightarrow$ $\frac{dw}{dx} + \frac{3}{x}w = \frac{3}{x}$

$P(x) = \frac{3}{x}$ $\rightarrow$ $e^{\int \frac{3}{x}dx} = x^3$ $\rightarrow$ $x^3\frac{dw}{dx} + x^2w = 3x^2$

$\rightarrow \frac{d}{dx}[x^3w] = 3x^2$

$\rightarrow x^3w = x^3 + c \rightarrow y^3 = 1 + cx^{-3}$

$$\frac{dy}{dx} + P(x)y = f(x)y^n,$$

Bernoulli's equation

$$w = y^{1-n} \quad \rightarrow dw = (1-n)y^{-n}dy \rightarrow \cdots$$

16. $\dfrac{dy}{dx} - y = e^x y^2$ $\qquad \rightarrow y^{-2}\dfrac{dy}{dx} - y^{-1} = e^x$

$$w = y^{1-2} = y^{-1} \rightarrow \frac{dw}{dx} = -1y^{-2}\frac{dy}{dx} \rightarrow -\frac{dw}{dx} - w = e^x \rightarrow \frac{dw}{dx} + w = -e^x$$

$$P(x) = +1 \rightarrow e^{\int dx} = e^x \rightarrow e^x\frac{dw}{dx} + e^x w = -e^{2x} \rightarrow \frac{d}{dx}[e^x w] = e^{2x}$$

$$\rightarrow e^x w = -\tfrac{1}{2}e^{2x} + c \rightarrow w = -\tfrac{1}{2}e^x + ce^{-x}$$

$$\rightarrow y^{-1} = -\frac{1}{2}e^x + ce^{-x}$$

17. $\frac{dy}{dx} = y(xy^3 - 1) \rightarrow \frac{dy}{dx} + y = xy^4 \rightarrow y^{-4}\frac{dy}{dx} + y^{-3} = x$

$w = y^{1-4} = y^{-3}$ $\rightarrow \frac{dw}{dx} = -3y^{-4}\frac{dy}{dx} \rightarrow \frac{dw}{dx} - 3\,w = -3x$

$P(x) = -3 \rightarrow e^{-3\int dx} = e^{-3x} \rightarrow e^{-3x}\frac{dw}{dx} - 3e^{-3x}w = -3xe^{-3x}$

$\rightarrow \frac{d}{dx}[e^{-3x}w] = -3xe^{-3x}$

$\rightarrow e^{-3x}w = xe^{-3x} + \frac{1}{3}e^{-3x} + c \rightarrow w = x + \frac{1}{3}e^{x} + ce^{-3x}$

$\rightarrow y^{-3} = x + \frac{1}{3}e^{x} + ce^{-3x}$

In Problems 21 and 22 solve the given initial-value problem.

21. $x^2 \dfrac{dy}{dx} - 2xy = 3y^4, \quad y(1) = \frac{1}{2}$

22. $y^{1/2} \dfrac{dy}{dx} + y^{3/2} = 1, \quad y(0) = 4$

21. $x^2 \dfrac{dy}{dx} - 2xy = 3y^4, \quad y(1) = \frac{1}{2} \quad \rightarrow \quad \dfrac{dy}{dx} - \dfrac{2}{x} y = \dfrac{3}{x^2} y^4 \quad \rightarrow \quad y^{-4} \dfrac{dy}{dx} - \dfrac{2}{x} y^{-3} = \dfrac{3}{x^2}$

$w = y^{1-4} = y^{-3} \quad \rightarrow \quad \dfrac{dw}{dx} = -3y^{-4} \dfrac{dy}{dx} \quad \rightarrow \quad \dfrac{dw}{dx} + \dfrac{6}{x} w = \dfrac{-9}{x^2}$

$P(x) = \dfrac{6}{x} \quad \rightarrow \quad e^{6 \int \frac{dx}{x}} = x^6 \quad \rightarrow \quad x^6 \dfrac{dw}{dx} + 6x^5 w = -9x^4$

$\rightarrow \dfrac{d}{dx}[x^6 w] = -9x^4$

$\rightarrow x^6 w = -\frac{9}{5} x^5 + c \quad \rightarrow \quad w = -\frac{9}{5} x^{-1} + cx^{-6}$

$\rightarrow \quad y^{-3} = -\frac{9}{5} x^{-1} + cx^{-6}$

$y(1) = \dfrac{1}{2} \rightarrow 8 = -\dfrac{9}{5} + c \rightarrow c = -\dfrac{49}{5} \quad \rightarrow y^{-3} = -\dfrac{9}{5} x^{-1} - \dfrac{49}{5} x^{-6}$

$$22.\quad y^{1/2}\frac{dy}{dx} + y^{3/2} = 1, \quad y(0) = 4 \quad \rightarrow \quad \frac{dy}{dx} + y = y^{\frac{-1}{2}} \quad \rightarrow \quad y^{\frac{1}{2}}\frac{dy}{dx} + y^{\frac{3}{2}} = 1$$

$$w = y^{1-\frac{-1}{2}} = y^{\frac{3}{2}} \quad \rightarrow \quad \frac{dw}{dx} = \frac{3}{2}y^{\frac{1}{2}}\frac{dy}{dx} \quad \rightarrow \quad \frac{dw}{dx} + \frac{3}{2}w = \frac{3}{2}$$

$$P(x) = \frac{3}{2} \quad \rightarrow \quad e^{\int \frac{3dx}{2}} = e^{\frac{3}{2}x} \quad \rightarrow e^{\frac{3}{2}x}\frac{dw}{dx} + \frac{3}{2}e^{\frac{3}{2}x}w = \frac{3}{2}e^{\frac{3}{2}x}$$

$$\rightarrow \frac{d}{dx}\left[e^{\frac{3}{2}x}w\right] = \frac{3}{2}e^{\frac{3}{2}x}$$

$$\rightarrow e^{\frac{3}{2}x}w = e^{\frac{3}{2}x} + c \quad \rightarrow \quad w = 1 + c\,e^{-\frac{3}{2}x}$$

$$\rightarrow y^{\frac{3}{2}} = 1 + c\,e^{-\frac{3}{2}x}$$

$$y(0) = 4 \rightarrow \quad 8 = 1 + c \ \rightarrow c = 7 \quad \rightarrow y^{-3} = 1 + 7\,e^{-\frac{3}{2}x}$$

In Problems 23–28 solve the given differential equation by using an appropriate substitution.

23. $\dfrac{dy}{dx} = (x + y + 1)^2$

24. $\dfrac{dy}{dx} = \dfrac{1 - x - y}{x + y}$

25. $\dfrac{dy}{dx} = \tan^2(x + y)$

26. $\dfrac{dy}{dx} = \sin(x + y)$

27. $\dfrac{dy}{dx} = 2 + \sqrt{y - 2x + 3}$

28. $\dfrac{dy}{dx} = 1 + e^{y-x+5}$

23. $\frac{dy}{dx} = (x + y + 1)^2$

$$u = x + y + 1$$

$$u = x + y + 1 \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$$

$$\frac{du}{dx} - 1 = u^2 \rightarrow \frac{1}{1+u^2} du = dx$$

$$\tan^{-1} u = x + c$$

$$u = \tan(x + c)$$

$$x + y + 1 = \tan(x + c)$$

$$y = \tan(x + c) - x - 1$$

24. $\dfrac{dy}{dx} = \dfrac{1 - x - y}{x + y}$

$u = x + y \rightarrow$

$$u = x + y \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$$

$$\frac{du}{dx} - 1 = \frac{1-u}{u} \rightarrow u\,du = dx$$

$$\frac{1}{2}u^2 = x + c$$

$$u^2 = 2x + c_1$$

$$(x + y)^2 = 2x + c_1$$

In Problems 29 and 30 solve the given initial-value problem.

29. $\dfrac{dy}{dx} = \cos(x + y), \quad y(0) = \pi/4$

30. $\dfrac{dy}{dx} = \dfrac{3x + 2y}{3x + 2y + 2}, \quad y(-1) = -1$

29. $$\frac{dy}{dx} = \cos(x+y), \quad y(0) = \pi/4$$

$$u = x + y$$

You may use the table

$$u = x + y \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$$

$$\frac{du}{dx} - 1 = \cos u \rightarrow \frac{1}{1+\cos u} du = dx$$

$$\frac{1}{1+\cos u} = \frac{1-\cos u}{1-\cos^2 u} = \frac{1-\cos u}{\sin^2 u} = \csc^2 u - \csc u \cot u$$

$$\int \left(\csc^2 u - \csc u \cot u\right) du = \int dx$$

$$-\cot u + \csc u = x + c$$

$$-\cot(x+y) + \csc(x+y) = x + c$$

$$x = 0, y = \frac{\pi}{4} \rightarrow c = \sqrt{2} - 1$$

$$\csc(x+y) - \cot(x+y) = x + \sqrt{2} - 1$$

30. $$\frac{dy}{dx} = \frac{3x + 2y}{3x + 2y + 2}, \quad y(-1) = -1$$

$$u = 3x + 2y$$

$$u = 3x + 2y \rightarrow \frac{du}{dx} = 3 + 2\frac{dy}{dx}$$

$$\frac{du}{dx} = 3 + \frac{2u}{u+2} = \frac{5u+6}{u+2} \rightarrow \frac{u+2}{5u+6}du = dx$$

$$\frac{u+2}{5u+6} = \frac{1}{5} + \frac{4}{25u+30}$$

$$\int \left(\frac{1}{5} + \frac{4}{25u+30}\right) du = \int dx$$

$$\frac{1}{5}u + \frac{4}{25}\ln|25u + 30| = x + c$$

$$\frac{1}{5}(3x + 2y) + \frac{4}{25}\ln|75x + 50y + 30| = x + c$$

$$x = -1, y = -1 \rightarrow c = \frac{4}{25}\ln 95$$

Replace c and simplify it $\longrightarrow$ $$5y - 5x + 2\ln|75x + 50y + 30| = 2\ln 95$$

Basic Integral Formulas

1. $\int u^n du = \frac{u^{n+1}}{n+1} + C$ (for $n \neq -1$)
2. $\int \frac{1}{u} du = \ln|u| + C$
3. $\int du = u + C$
4. $\int e^u du = e^u + C$
5. $\int a^u du = \frac{a^u}{\ln a} + C$ (for $a > 0, a \neq 1$)
6. $\int \ln u du = u \ln u - u + C$
7. $\int \sin u du = -\cos u + C$
8. $\int \cos u du = \sin u + C$
9. $\int \sin^2 u du = \frac{u}{2} - \frac{\sin 2u}{4} + C$
10. $\int \cos^2 u du = \frac{u}{2} + \frac{\sin 2u}{4} + C$
11. $\int \tan u du = -\ln|\cos u| + C = \ln|\sec u| + C$
12. $\int \sec u du = \ln|\sec u + \tan u| + C$
13. $\int \sec^2 u du = \tan u + C$
14. $\int \sec u \tan u du = \sec u + C$
15. $\int \cot u du = \ln|\sin u| + C = -\ln|\csc u| + C$
16. $\int \csc u du = \ln|\csc u - \cot u| + C$